MD. TASNIM AZAD

House-28, Road-2, Block-B, Banasree, Dhaka
Contact: 01724738012
E-mail: nihan.tas.2016@gmail.com
Linkedin: Tasnim Azad Nihan
Github: nihan139

PROFILE

An enthusiastic Electrical Engineer having an active interest in semiconductor based device design, VLSI circuits and Digital Logic Design. Currently I am working as an adjunct lecturer at Brac University. I want to enrich my knowledge and adapt to new environment and projects.

EDUCATION

Bangladesh University of Engineering & Technology — **May, 2023**

Bachelor of Science in Electrical and Electronic Engineering

- **MAJOR: Electronics**
- **CGPA: 3.93**
  (7th out of 189 students)

Notre Dame College — **2017**

Higher Secondary School Certificate

- **GPA: 5.00**

Ideal School and College — **2015**

Secondary School Certificate

- **GPA: 5.00**

SKILLS

Device Simulation	**Silvaco TCAD, KeySight ADS**
Cadence Tools	**Virtuoso, Genus, Innovus**
Circuit Simulation	**Proteus, PSpice, Tinkercad**
Microcontroller	**Arduino, ESP32**
Coding Skills	**Python, Matlab, LaTeX, C, C++, HTML**
Others	**Autocad, CYME PSAF**

TEACHING EXPERIENCE

Adjunct Lecturer, CSE, Brac University **September,23 - Present**

Courses Instructed:

- **Circuits and Electronics**
- **Electronic Devices and Circuits**

PUBLICATION

AlGaN/GaN based Multi-Metal Gated High Electron Mobility Transistor with Improved Linearity

(MIT Collaboration Work)

Supervisor(Dr. Nadim Chowdhury)

Published in IEEE Transactions on Electron Devices - *Link*

Developing a novel approach to improve the linearity of AlGaN-GaN HEMT structure which also focuses on to make the device compatible in high efficiency power amplifiers for 5G communication.

SPECIAL ACHIEVEMENT

- **Dean's List Award and University Scholarship in all 8 terms**
- Our designed **'Wifi-updated Bangla Calendar Clock'** is mounted on the wall at **'Microprocessor Lab, BUET'** for 6 months as a recognition of our work.

PROJECTS

- **Design of a RISC-V 32 Bit Processor** (Ongoing Work)
  (Very Large Scale Integrated Circuit)
- **High Speed Digital IO RX** - *Link*
  (Analog Integrated Circuit)
- **Design of a Low Cost Spectrometer** - *Link*
  (Optoelectronics)
- **Wi-Fi Updated Bangla Calendar Clock** - *Link*
  (Microcontroller Based Project)
- **Electrical Design of a Multi-storied Building** - *Link*
  (Autocad Project)
- **Real Time Object Detection for Blind People** - *Link*
  (Neural Network Model Based)
- **Automatic Hand Sanitization Booth & Temperature Controlled Door Unlocking System** - *Link*

(Control System Project)

- **Fire Alarm Circuit using custom 4 bit Analog to Digital Converter** - *Link*
  (Digital Logical Design)
- **Battery Charger with Variable DC Output Using a Buck-Boost Converter with Auto-Cutoff Feature** - *Link*
  (Simulink Project)
- **Designing a Communication System Based on Delta Modulation** - *Link*
  (Proteus Project)

COMPETITION PARTICIPATION

- **VLSI DAY 2022** - *Certificate Link*
  (Organized by the Department of EEE, United International University)
- **VLSITHON 1.0** - *Certificate Link*
  (Organized by Ulkasemi Limited)
- **VLSI DESIGN COMPETITION 2022** - *Certificate Link*
  (Jointly organized by the Department of EEE, BUET and IEEE ED/SSCS Bangladesh Chapter)

REFERENCE

- **Dr. Nadim Chowdhury**
  Assistant Professor, Department of Electrical and Electronic Engineering
  Bangladesh University of Engineering and Technology
- **Bejoy Sikder**
  Lecturer, Department of Electrical and Electronic Engineering
  Bangladesh University of Engineering and Technology